

18)Develop a Digital Clock and Examination Countdown Timer

HTML display elements, CSS styling, JavaScript Date object, setInterval(), functions and DOM updating.

CODE

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Digital Clock and Examination Countdown</title>
```

```
<style>
```

```
  * {
```

```
    box-sizing: border-box;
```

```
    margin: 0;
```

```
    padding: 0;
```

```
  }
```

```
  body {
```

```
    font-family: Arial, sans-serif;
```

```
    background: linear-gradient(135deg, #141e30, #243b55);
```

```
    min-height: 100vh;
```

```
    display: flex;
```

```
    justify-content: center;
```

```
    align-items: center;
```

```
    padding: 20px;
```

```
}
```

```
.container {  
  width: 100%;  
  max-width: 600px;  
  background: white;  
  padding: 30px;  
  border-radius: 20px;  
  text-align: center;  
  box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);  
}
```

```
h1 {  
  color: #243b55;  
  margin-bottom: 25px;  
}
```

```
.clock-box,  
.countdown-box {  
  padding: 25px;  
  margin: 15px 0;  
  border-radius: 15px;  
  background: #f2f5f8;  
}
```

```
h2 {  
  color: #243b55;  
  margin-bottom: 15px;
```

```
}
```

```
#clock {  
    font-size: 42px;  
    font-weight: bold;  
    color: #667eea;  
}
```

```
#date {  
    margin-top: 10px;  
    font-size: 18px;  
    color: #555;  
}
```

```
#countdown {  
    font-size: 35px;  
    font-weight: bold;  
    color: #e63946;  
    margin-top: 10px;  
}
```

```
#message {  
    margin-top: 15px;  
    font-size: 18px;  
    font-weight: bold;  
    color: #2a9d8f;  
}
```

```
@media (max-width: 500px) {  
  #clock {  
    font-size: 32px;  
  }  
  
  #countdown {  
    font-size: 27px;  
  }  
  
  .container {  
    padding: 20px;  
  }  
}  
</style>  
</head>  
  
<body>  
  
<div class="container">  
  
<h1><img alt="alarm clock icon" data-bbox="183 683 206 698"/> Examination Timer</h1>  
  
<div class="clock-box">  
  <h2>Digital Clock</h2>  
  <div id="clock">00:00:00</div>  
  <div id="date">Loading...</div>  
</div>
```

```
<div class="countdown-box">

  <h2>Examination Countdown</h2>

  <div id="countdown">Loading...</div>

  <div id="message"></div>

</div>
```

```
</div>
```

```
<script>
```

```
function updateClock() {

  const now = new Date();


  let hours = now.getHours();

  let minutes = now.getMinutes();

  let seconds = now.getSeconds();


  hours = hours < 10 ? "0" + hours : hours;

  minutes = minutes < 10 ? "0" + minutes : minutes;

  seconds = seconds < 10 ? "0" + seconds : seconds;


  document.getElementById("clock").innerHTML =

    hours + ":" + minutes + ":" + seconds;


  document.getElementById("date").innerHTML =

    now.toString();

}
```

```
function updateCountdown() {
```

```
const now = new Date();
```

```
const examDate = new Date();
```

```
examDate.setHours(23, 59, 59, 0);
```

```
const difference = examDate - now;
```

```
if (difference <= 0) {
```

```
    document.getElementById("countdown").innerHTML = "00:00:00";
```

```
    document.getElementById("message").innerHTML =
```

```
        "Examination Time Reached!";
```

```
    return;
```

```
}
```

```
const hours = Math.floor(difference / (1000 * 60 * 60));
```

```
const minutes = Math.floor(
```

```
    (difference % (1000 * 60 * 60)) / (1000 * 60)
```

```
);
```

```
const seconds = Math.floor(
```

```
    (difference % (1000 * 60)) / 1000
```

```
);
```

```
document.getElementById("countdown").innerHTML =
```

```
    String(hours).padStart(2, "0") + ":" +
```

```
    String(minutes).padStart(2, "0") + ":" +
```

```
    String(seconds).padStart(2, "0");
```

```
document.getElementById("message").innerHTML =
```

```
        "Time remaining for examination";
    }

    updateClock();
    updateCountdown();

    setInterval(updateClock, 1000);
    setInterval(updateCountdown, 1000);
</script>

</body>
</html>
```

OUTPUT

